

Low-Level Design Document

ETL7 — Low-Level Design Document

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Field	Value
Cluster	etl7
Role	Managed (OpenShift Virtualization workload cluster)
OCP Version	4.22.8 (stable-4.22 channel)
Date	August 15, 2026
Generated From	GitOps repository — virtualization-migration-factory
Domain	etl7.ocp.rht-labs.com

Executive Summary

The **etl7** cluster is a compact, bare-metal OpenShift Virtualization workload cluster running Red Hat OpenShift Container Platform 4.22. It operates as an ACM-managed cluster within a multi-cluster fleet governed by GitOps (ArgoCD) from a centralized hub cluster.

The cluster is purpose-built for running virtual machine workloads using OpenShift Virtualization (KubeVirt), backed by NetApp ONTAP-SAN block storage via Astra Trident. With 3 converged master/worker nodes (HPE ProLiant DL380 Gen9), it runs a full production-grade operator stack including high-availability features (Node Health Check, Descheduler), comprehensive observability (Prometheus, Loki, Grafana, OpenTelemetry, Network Observability), and policy enforcement (Kyverno).

All cluster configuration is declared in Git and delivered through ArgoCD using the app-of-apps pattern. Operator lifecycle management is handled via ACM OperatorPolicy resources, providing centralized governance over operator installation, upgrade channels, and version pinning.

Multi-Cluster Architecture

The etl7 cluster is part of a 4-cluster fleet managed by a centralized hub:

Cluster	Role	Groups	Nodes
hub	Hub (ACM + ArgoCD control plane)	all	—
etl4	Managed (workload)	all, prod	dl380g9-1, dl380g9-2, dl380g9-3
etl6	Managed (workload)	all, prod	dl380g9-5, dl380g9-6, dl380g9-7
etl7	Managed (workload)	all, prod	dl380g9-8, dl380g9-9, dl380g9-10

Fleet Governance Model

- **Hub cluster** runs Red Hat Advanced Cluster Management (ACM) and ArgoCD, provisioning and managing all spoke clusters.
- **GitOps delivery:** The hub's ArgoCD instance renders each cluster's configuration from the shared Git repository using Kustomize + Helm, delivered through a custom `envsubst` CMP sidecar.
- **Policy-based lifecycle:** Operators are deployed and upgraded via ACM `operatorPolicy` resources — not traditional OLM Subscriptions — enabling centralized version control and upgrade approval across the fleet.
- **Monitoring federation:** Prometheus on `etl7` forwards alerts to the hub's ACM Observability Alertmanager (`alertmanager-open-cluster-management-observability.apps.hub2.ocp.rht-labs.com`).

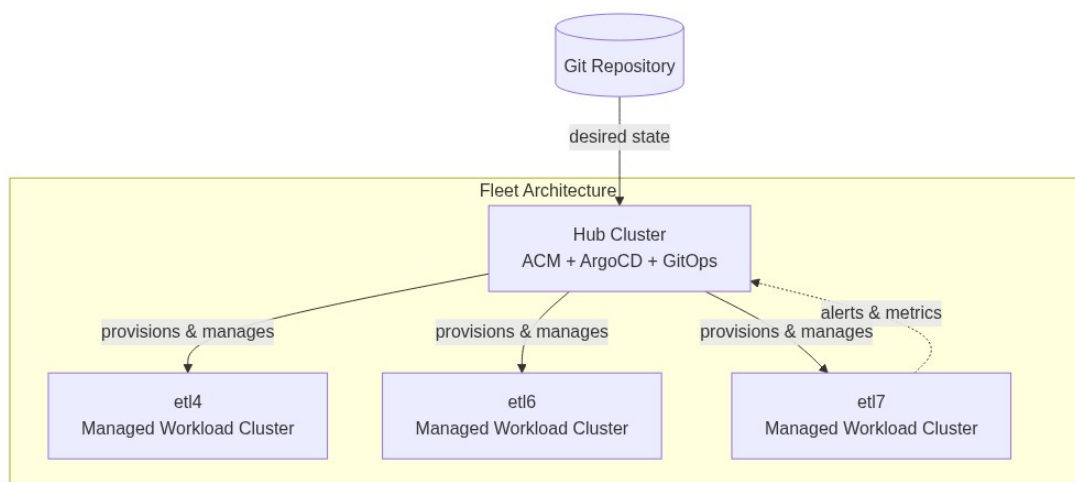


Diagram 1

Cluster Identity

Property	Value
Name	etl7
Role	Managed (spoke)
OCP Version	4.22.8
Update Channel	stable-4.22
Git Version Pin	main
Domain	etl7.ocp.rht-labs.com
Groups	<code>all</code> (base infrastructure), <code>prod</code> (virtualization + HA + observability)
Active Components	48
Commented-Out (Available)	Additional components available but not enabled

Group Contributions

Group	What it provides
all	Base infrastructure: GitOps operator, cert-manager, external-dns, MetalLB, NMState, Trident storage, Loki logging, MinIO (log storage), monitoring configuration, kube-ops-view
prod	Virtualization platform: OpenShift Virtualization, Descheduler, Node Health Check, Kyverno policies, Grafana, Network Observability, OpenTelemetry, Tempo, ScyllaDB, Cluster Observability

Compute & Node Inventory

Node Hardware

All nodes are **HPE ProLiant DL380 Gen9** rack servers operating in a **compact/converged configuration** — all 3 nodes serve as both control plane (master) and worker nodes. This means the cluster does not have dedicated infrastructure nodes; platform services and workloads share the same compute.

Hostname	FQDN	Role	BMC Address	Boot MAC	Status
dl380g9-8	dl380g9-8.etl7.ocp.rht-labs.com	master	redfish://10.9.48.218	00:11:0a:6b:b5:40	Online
dl380g9-9	dl380g9-9.etl7.ocp.rht-labs.com	master	redfish://10.9.48.219	00:11:0a:6a:65:b8	Online
dl380g9-10	dl380g9-10.etl7.ocp.rht-labs.com	master	redfish://10.9.48.220	00:11:0a:68:06:4c	Online

NIC Inventory (per node)

Interface	Type	MAC Address (dl380g9-8)	State	Purpose
ens2f0	Ethernet	00:11:0a:6b:b5:40	up	Bond member (primary)
ens2f1	Ethernet	00:11:0a:6b:b5:41	up	Bond member
eno1	Ethernet	30:e1:71:6f:c8:30	up	Reserved/unused
eno2	Ethernet	30:e1:71:6f:c8:31	down	Unused
eno3	Ethernet	30:e1:71:6f:c8:32	down	Unused
eno4	Ethernet	30:e1:71:6f:c8:33	down	Unused

Compute Notes

- **Masters schedulable:** All 3 nodes carry the `master` role in a compact 3-node deployment, meaning virtual machines are scheduled directly on control plane nodes.
- **No dedicated workers:** This is appropriate for lab/development environments. For production workloads requiring strict separation, dedicated worker nodes should be considered.

- **Live migration:** Supported across all 3 nodes since they share identical hardware (same HPE DL380 Gen9 generation, same CPU model).

Networking

6.1 Installation Network

The installation-time network uses bonded interfaces with tagged VLANs for control plane and storage traffic separation.

Parameter	Value
Bond Mode	802.3ad (LACP)
Bond Members	ens2f0, ens2f1
MII Monitor	100ms
Control Plane VLAN	2531
Control Plane Subnet	10.9.53.0/25
Default Gateway	10.9.53.1 (via bond0.2531)
Storage VLAN	3531
Storage Subnet	192.168.53.0/25
DNS Servers	10.9.48.31, 10.9.48.32
IP Assignment	Static

Per-Node IP Addressing

Node	Control Plane IP	Storage IP
dl380g9-8	10.9.53.18/25	192.168.53.18/25
dl380g9-9	10.9.53.19/25	192.168.53.19/25
dl380g9-10	10.9.53.20/25	192.168.53.20/25

6.2 Day-2 Network Configuration

Post-installation, a dedicated OVS bridge is configured for VM traffic, separate from the cluster SDN:

Parameter	Value
Bridge Name	ovs-br1
Bridge Type	OVS (Open vSwitch)
Port	bond0
STP	Disabled
Node Selector	<code>node-role.kubernetes.io/worker: ''</code> (all nodes)

OVN Bridge Mappings

Localnet Name	Bridge	State	Purpose
vlan-2522	ovs-br1	present	VM network traffic
vlan-2901	ovs-br1	present	VM secondary network (192.168.90.0/24)
vlan-2911	ovs-br1	present	VM tertiary network (192.168.91.0/24)

Network Attachment Definitions (NADs)

NAD Name	VLAN ID	Type	Namespace	MTU
vlan-2522	2522	ovn-k8s-cni-overlay (localnet)	default	1500
vlan-2901	2901	ovn-k8s-cni-overlay (localnet)	default	1500
vlan-2911	2911	ovn-k8s-cni-overlay (localnet)	default	1500

6.3 Load Balancing (MetalLB)

Parameter	Value
IP Address Pool	main-pool
Address Range	10.9.53.32/28 (16 addresses: .32–.47)
Mode	L2 (Layer 2 advertisement)

6.4 Cluster Network Configuration

Parameter	Value
SDN	OVN-Kubernetes
Route Advertisements	Disabled
FRR (BGP)	Available but not enabled (commented out)

6.5 Network Topology Diagram

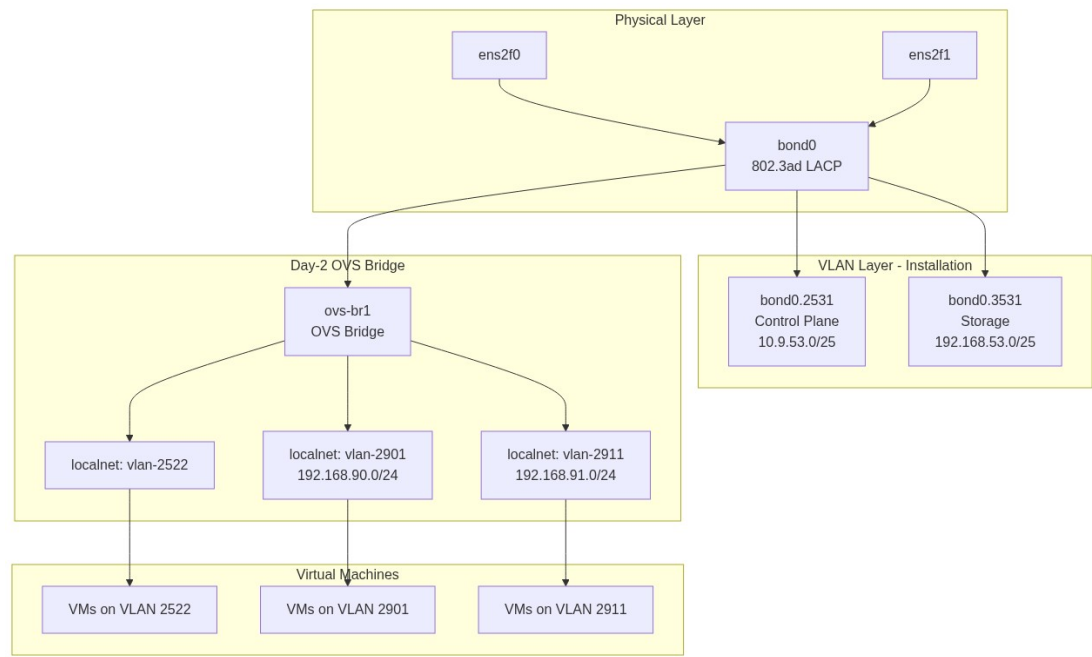


Diagram 2

6.6 Available but Inactive (Commented-Out)

- **Infinidat iSCSI storage network:** Per-node Linux bridges on VLAN 2911 for dedicated iSCSI traffic to Infinidat storage (IPs: 192.168.91.8–.10). Currently disabled — available for future Infinidat integration.

Storage

Storage Backend

Parameter	Value
Vendor	NetApp
Product	ONTAP
Driver	ontap-san (iSCSI block)
CSI Provider	Astra Trident (csi.trident.netapp.io)
Management LIF	netapp.etl.rht-labs.com
SVM	ocp_e7
Credentials	Managed via Kubernetes Secret (ontap-san-secret)

Storage Classes

StorageClass	Provisioner	Backend	Reclaim	Expansion	Default Virt	Binding
ontap-san	csi.trident.netapp.io	ontap-san	Delete	Yes	Yes	Immediate
ontap-nas	csi.trident.netapp.io	ontap-nas	Delete	Yes	No	Immediate

Storage Configuration Notes

- **ontap-san** is the default virtualization StorageClass (`storageclass.kubevirt.io/is-default-virt-class: "true"`), meaning all VM disks are provisioned as iSCSI LUNs by default.
- Thin provisioning is enabled, with snapshot and clone support.
- **ontap-nas** is used for infrastructure workloads requiring ReadWriteMany (RWX) access — notably Prometheus persistent storage (100Gi).
- iSCSI block volumes inherently support RWX access mode, enabling VM live migration without additional configuration.

Installation Method

Parameter	Value
Method	ACM Agent-Based Install
Provisioning Hub	hub cluster
BMC Protocol	Redfish (HPE iLO)
Boot Method	Virtual Media (Redfish)
Automated Cleaning	Metadata only
Infrastructure	BareMetalHost + InfraEnv + AgentClusterInstall

How It Works

1. The hub cluster's ACM Multi-Cluster Engine (MCE) provisions etl7 via agent-based installation.
2. `BareMetalHost` custom resources define each node's BMC address and boot MAC.
3. MCE uses Redfish Virtual Media to boot the discovery ISO on each node — no PXE infrastructure required.
4. `NMStateConfig` resources configure static networking (bonds, VLANs, IPs) during installation.
5. Once agents register, `AgentClusterInstall` drives the cluster deployment to completion.
6. Post-install, ArgoCD applies the full operator stack and day-2 configuration from Git.

Operator Stack

All operators are deployed via ACM **OperatorPolicy** resources (not traditional Subscriptions). This provides centralized governance, version pinning, and fleet-wide upgrade coordination.

Wave 5 — Operators

Operator	Package	Channel	Source	Namespace	Starting CSV	Upgrade
cert-manager	cert-manager	stable-v1	certified-operators	cert-manager	cert-manager.v1.16.0	Automatic
cluster-observability	—	—	—	—	—	—
descheduler	cluster-kube-descheduler-operator	stable	redhat-operators	openshift-kube-descheduler-operator	v5.1.1	Automatic
external-dns	external-dns-operator	stable-v1	redhat-operators	external-dns-operator	v1.3.0	Automatic
kyverno	—	—	—	—	—	—
loki	loki-operator	stable-6.6	redhat-operators	openshift-loki-operator	v6.3.0	Automatic
metallb	metallb-operator	stable	redhat-operators	metallb-system	v4.20.0	Automatic
minio	—	—	—	—	—	—
network-observability	netobserv-operator	stable	redhat-operators	openshift-netobserv-operator	v1.12.0	Automatic
nmstate	kubernetes-nmstate-operator	stable	redhat-operators	openshift-nmstate	v4.19.0	Automatic
node-health-check	—	—	—	—	—	—
openshift-gitops	openshift-gitops-operator	latest	redhat-operators	openshift-gitops-operator	v1.20.4	Automatic
openshift-grafana	—	—	—	—	—	—
openshift-logging	cluster-logging	stable-6.6	redhat-operators	openshift-logging	v6.3.0	Automatic
openshift-tempo	—	—	—	—	—	—
openshift-virtualization	kubevirt-hyperconverged	stable	redhat-operators	openshift-cnv	v4.20.11	Automatic
otel	—	—	—	—	—	—
scylladb	scylladb-operator	stable	certified-operators	scylladb	v1.20.2	Automatic
trident	trident-operator	stable	certified-operators	netapp-trident	v25.6.1	Automatic

Wave 6 — Early Dependencies

Component	Purpose
cert-manager-configuration	ClusterIssuers and certificates for TLS
external-dns-configuration	External DNS record management
metallb-configuration	IP address pool (10.9.53.32/28)
nmstate-instance	NMState operator instance
trident-configuration	ONTAP backend + StorageClass (cluster-specific)
trident-instance	Trident orchestrator instance

Wave 15 — Instances & Configurations

Component	Purpose
cluster-observability-configuration	Observability stack configuration
descheduler-configuration	VM workload rebalancing policies
hyperconverged-instance	OpenShift Virtualization HyperConverged CR
kube-ops-view	Cluster visualization dashboard
kyverno-configuration	Security and compliance policies
loki-logging-configuration	Log aggregation configuration
network-observability-configuration	Network flow monitoring
nmstate-configuration	Day-2 OVS bridges and bridge mappings (cluster-specific)
node-health-check-configuration	Node self-healing policies
openshift-config	OAuth, console, monitoring, network patches (cluster-specific)
openshift-grafana-instance	Grafana dashboards
openshift-logging-instance	Log collection and forwarding
openshift-tempo-instance	Distributed tracing backend
reflector-operator	Cross-namespace ConfigMap/Secret propagation
scylladb-instance	ScyllaDB cluster instance
user-workload-monitoring	User application metrics collection
wasp-agent	VirtIO-fs cache agent for VM performance

Wave 25 — Applications

Component	Purpose
openshift-grafana-application	Grafana datasource and dashboard definitions

Observability

Monitoring Stack

The etl7 cluster runs a comprehensive observability stack:

Component	Technology	Purpose
Platform Monitoring	Prometheus + Alertmanager	Core cluster metrics and alerting
User Workload Monitoring	Prometheus (user namespaces)	Application-level metrics
Logging	Loki + OpenShift Logging	Centralized log aggregation
Log Storage	MinIO	S3-compatible object storage for Loki
Dashboards	Grafana	Custom visualization and dashboards
Network Observability	NetObserv (eBPF)	Network flow analysis and monitoring
Distributed Tracing	Tempo + OpenTelemetry	Request tracing across services
Cluster Observability	Cluster Observability Operator	Consolidated observability management

Prometheus Configuration

Parameter	Value
User Workload Monitoring	Enabled
Prometheus Memory Limit	32Gi
Prometheus Memory Request	8Gi
Prometheus Storage	100Gi (ontap-nas StorageClass)
Alertmanager Federation	Hub ACM Observability
External Label	managed_cluster: 57ad5a67-2446-457c-992b-a3e05269b7ec
Node Exporter Collectors	buddyinfo, cpufreq, ksmd, mountstats, netclass, netdev, processes, systemd, tcpstat

Console Plugins

The OpenShift web console is extended with the following plugins:

Plugin	Purpose
monitoring-plugin	Enhanced monitoring UI
gitops-plugin	ArgoCD integration
kubevirt-plugin	VM management
nmstate-console-plugin	Network state visibility
node-remediation-console-plugin	Node health actions
networking-console-plugin	Network configuration
logging-view-plugin	Log viewing
soteria-console-plugin	Security insights
netobserv-plugin	Network flow visualization

Authentication & Authorization

Provider	Type	Method
htpasswd	HTPasswd	File-based (htpass-secret)

Notes

- The cluster currently uses **HTPasswd** as its sole identity provider. This is typical for lab/development environments.
- For production readiness, integration with an enterprise identity provider (OIDC/Okta, LDAP, or Active Directory) is recommended.
- The htpasswd secret is managed as a Kubernetes Secret referenced by the OAuth CR.
- ArgoCD sync options include `Delete=false` to prevent accidental removal of the OAuth configuration.

Cluster Upgrades

OpenShift Platform

Parameter	Value
Current Version	4.22.8
Update Channel	stable-4.22
Desired Update	4.22.8 (current)
Strategy	Declarative via GitOps (ClusterVersion CR in Git)

Operator Upgrade Policy

All operators on this cluster are configured with **Automatic** upgrade approval, meaning they will upgrade within their approved version lists without manual intervention.

Operator	Upgrade Approval	Version Range
OpenShift Virtualization	Automatic	4.17.4 → 4.22.2
Trident	Automatic	v25.6.1 → v26.6.0
MetalLB	Automatic	4.18.0 → 4.22.0
NMState	Automatic	4.17.0 → 4.22.0
Loki	Automatic	v6.1.1 → v6.6.0
Logging	Automatic	v6.1.1 → v6.6.0
Descheduler	Automatic	v5.1.1 → v5.4.0
External DNS	Automatic	v1.3.0 → v1.3.8
Network Observability	Automatic	v1.12.0 → v1.12.1
GitOps	Automatic	v1.20.4 → v1.21.1
ScyllaDB	Automatic	v1.20.2 (pinned)

Upgrade Coordination

Upgrades are managed declaratively: 1. **Platform upgrades** are triggered by updating the `ClusterVersion` CR's `desiredUpdate.version` in Git. 2. **Operator upgrades** are controlled by the `versions[]` list in each `OperatorPolicy` — only versions in the list are approved. 3. The hub cluster's ACM governance ensures upgrade policies are applied consistently across the fleet.

Component Deployment Sequence

The complete component stack deploys in sync-wave order via ArgoCD:

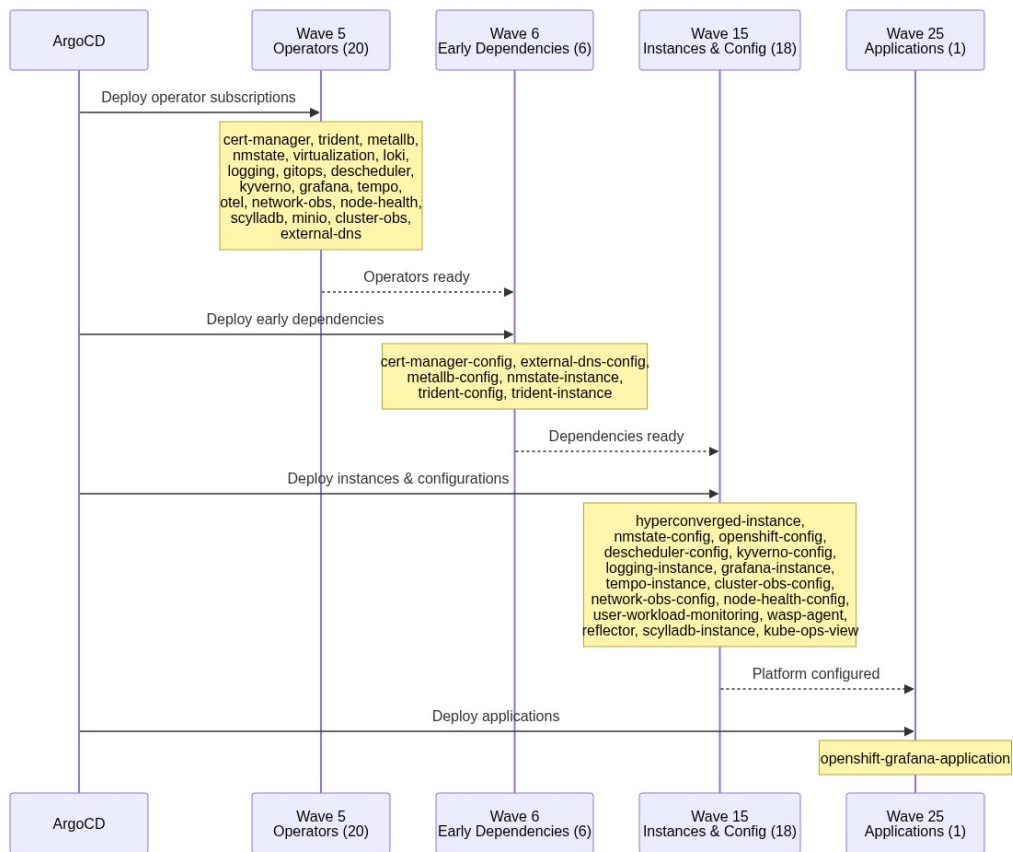


Diagram 3

Full Component List

#	Component	Wave	Source	Has Overlay
1	cert-manager-operator	5	group:all	No
2	cluster-observability-operator	5	group:all	No
3	descheduler-operator	5	group:prod	No
4	external-dns-operator	5	group:all	No
5	kyverno-operator	5	group:prod	No
6	loki-operator	5	group:all	No
7	metallb-operator	5	group:all	No
8	minio-operator	5	group:all	No
9	network-observability-operator	5	group:prod	No
10	nmstate-operator	5	group:all	No
11	node-health-check-operator	5	group:prod	No
12	openshift-gitops-operator	5	group:all	No
13	openshift-grafana-operator	5	group:prod	No
14	openshift-logging-operator	5	group:all	No
15	openshift-tempo-operator	5	group:prod	No
16	openshift-virtualization	5	group:prod	No
17	otel-configuration	5	group:prod	No
18	otel-operator	5	group:prod	No
19	scylladb-operator	5	group:prod	No
20	trident-operator	5	group:all	No
21	cert-manager-configuration	6	group:all	No
22	external-dns-configuration	6	group:all	No
23	metallb-configuration	6	cluster-specific	Yes
24	nmstate-instance	6	group:all	No
25	trident-configuration	6	cluster-specific	Yes
26	trident-instance	6	group:all	No
27	cluster-observability-configuration	15	group:prod	No
28	descheduler-configuration	15	group:prod	No
29	hyperconverged-instance	15	group:prod	No
30	kube-ops-view	15	group:all	No
31	kyverno-configuration	15	group:prod	No
32	loki-logging-configuration	15	group:all	No
33	network-observability-configuration	15	group:prod	No
34	nmstate-configuration	15	cluster-specific	Yes

#	Component	Wave	Source	Has Overlay
35	node-health-check-configuration	15	group:prod	No
36	openshift-config	15	cluster-specific	Yes
37	openshift-grafana-instance	15	group:prod	No
38	openshift-logging-instance	15	group:all	No
39	openshift-tempo-instance	15	group:prod	No
40	reflector-operator	15	group:all	No
41	scylladb-instance	15	group:prod	No
42	user-workload-monitoring	15	group:prod	No
43	wasp-agent	15	group:prod	No
44	openshift-grafana-application	25	group:prod	No

Design Decisions

ID	Topic	Decision	Rationale
DD-01	Cluster Topology	Compact 3-node (converged master/worker)	Suitable for lab/pre-production with moderate VM density; reduces hardware footprint while maintaining HA quorum
DD-02	Storage Backend	NetApp ONTAP-SAN (iSCSI) via Trident	Block storage provides RWX for live migration, thin provisioning reduces waste, enterprise-grade with snapshots and clones
DD-03	VM Network Isolation	Dedicated OVS bridge (ovs-br1) separate from cluster SDN	Prevents VM traffic from impacting control plane; allows independent VLAN management without risking SDN stability
DD-04	Network Attachment	OVN-Kubernetes localnet topology	Platform-level network policy support; better security controls vs raw Linux bridge; native VLAN tagging
DD-05	Identity Provider	HTPasswd	Appropriate for lab environment; production should integrate OIDC/Okta for enterprise SSO
DD-06	Operator Lifecycle	ACM OperatorPolicy (not OLM Subscription)	Fleet-wide governance, centralized version control, explicit version approval lists prevent unplanned upgrades
DD-07	Upgrade Strategy	Automatic approval within version lists	Balances automation with safety — operators auto-upgrade but only to pre-approved versions
DD-08	Update Channel	stable-4.22	Conservative channel; receives only bug fixes and security patches, not new features
DD-09	Monitoring Retention	100Gi Prometheus PVC on ontap-nas	Adequate for multi-month metrics retention in a 3-node lab cluster
DD-10	Alert Federation	Forward to hub ACM Observability	Centralized alerting enables fleet-wide incident response from a single pane of glass
DD-11	Load Balancing	MetalLB L2 with /28 pool	Simple L2 mode appropriate for flat network topology; 16 IPs sufficient for services in a lab cluster
DD-12	Node Health	Node Health Check operator	Automated node remediation reduces mean-time-to-recovery for hardware or OS failures

References

- [Red Hat OpenShift Container Platform 4.22 Documentation](#)
- [OpenShift Virtualization 4.22](#)
- [Red Hat Advanced Cluster Management for Kubernetes](#)
- [NetApp Astra Trident Documentation](#)
- [Kubernetes NMState Operator](#)
- [MetalLB Operator](#)
- [OpenShift Logging \(Loki\)](#)
- [OpenShift Monitoring](#)
- [cert-manager on OpenShift](#)
- [Kyverno Policy Engine](#)
- [ScyllaDB Operator](#)